

Project

1. Introduction & Research Question

Research Motivation

The topological structure of human online social networks such as Facebook does not emerge randomly. The classic Erdos - Renyi random graph assumes that the probability of forming a connection between any pair of individuals is identical. However, this assumption contradicts the fundamental social principle that "birds of a feather flock together".

Core Research Question

Which classical network generative model — the Erdos - Renyi random graph model, the Watts - Strogatz small-world model, or the Barabasi - Albert preferential attachment model — best reproduces the macroscopic and microscopic topological properties of the real-world Facebook network?

Hypothesis

We hypothesize that the empirical Facebook network simultaneously exhibits two key features: high local clustering (a hallmark of small-world networks) and heavy-tailed heterogeneity in its degree distribution (the scale-free property). This will invalidate the simple random graph as an adequate representation of real social connections.

2. Methodology

To rigorously evaluate the suitability of candidate models, we extract structural metrics from the empirical Facebook network and compare them against synthetic networks generated with identical scale parameters

($N = 4,039$ and $M = 88,234$, yielding an average degree $\langle k \rangle \approx 43.7$).

2.1 Dataset Selection

We adopt the Stanford SNAP Facebook Ego-Networks dataset, which contains $N=4039$ nodes and $E=88234$ edges.

2.2 Analytical Framework

Clustering Coefficient & Triadic Closure:

We measure the global clustering coefficient C and average local clustering coefficient to quantify the prevalence of closed triangles ("friends of my friends are my friends").

Degree Distribution & Power-Law Verification:

We compute the empirical degree distribution $P(k)$ and test whether it follows a power-law tail $P(k) \sim k^{-\gamma}$, characteristic of scale-free networks driven by preferential attachment. We examine whether the distribution exhibits power-law tail behaviour on a log-log scale.

Path Lengths & Small-World Property:

We compute the network diameter and average shortest path length L to evaluate the small-world phenomenon ($L \sim \ln(N)$) combined with high clustering.

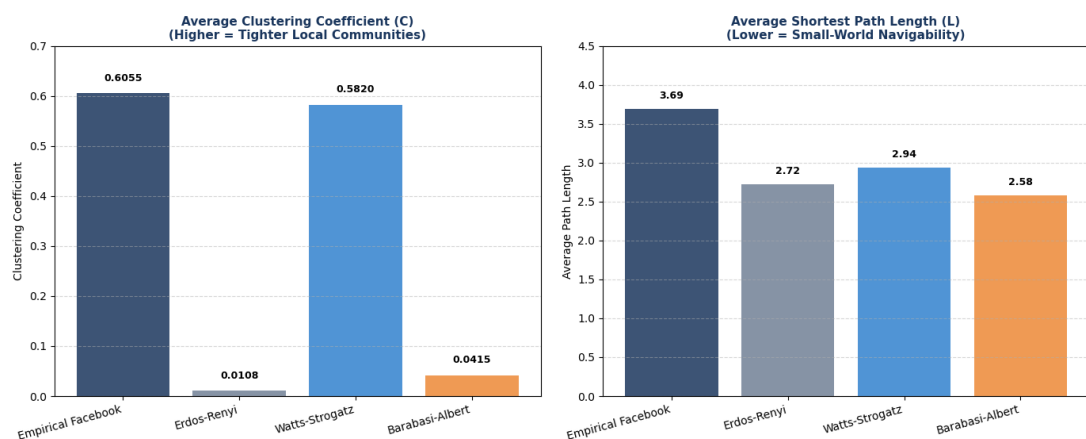
Null Model Benchmarking:

We construct baseline Erdos-Renyi $G(n, p)$ graphs, Watts-Strogatz $WS(n, k, p)$

models, and Barabasi-Albert $BA(n, m)$ models under controlled parameter constraints.

3. Results

3.1: Comparison of Macroscopic Topological Metrics (Corresponding to Figure 1: Dual-panel Metric Comparison)

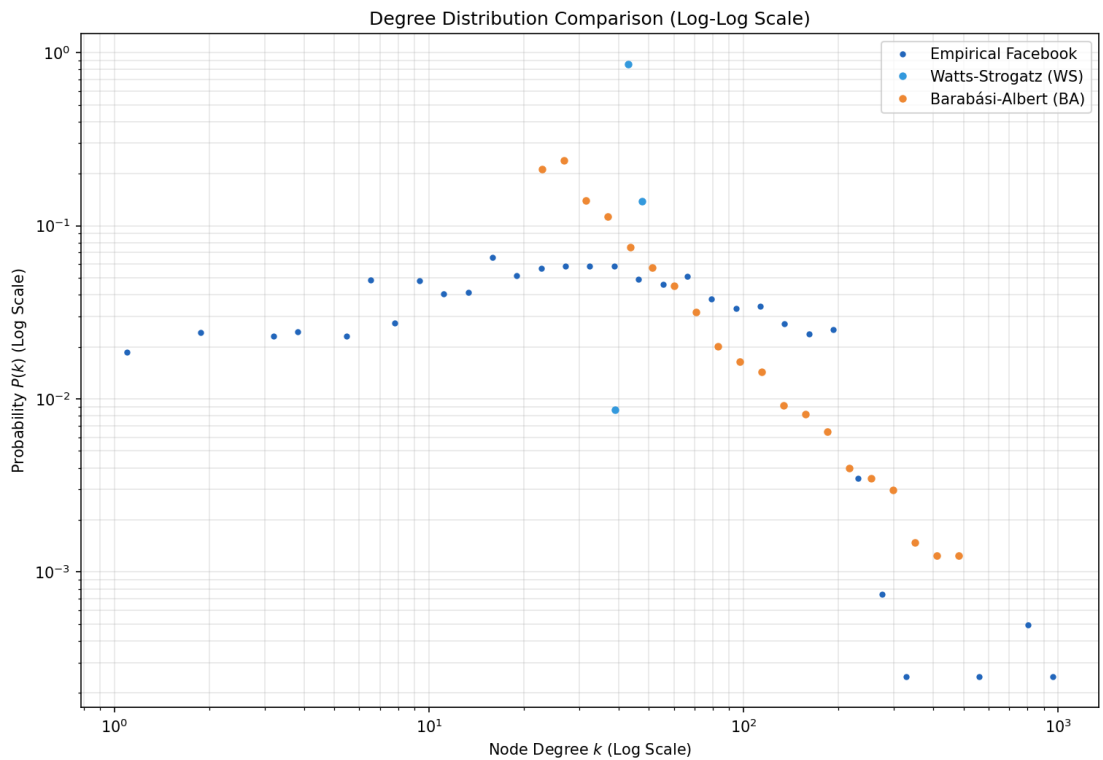


The empirical Facebook network achieves an average clustering coefficient of $C=0.6055$. This figure is far larger than the value of the Erdős-Rényi random graph (0.0108) and closely matches the result

generated by the Watts - Strogatz small-world model (0.5820). For average shortest path length, the real-world Facebook network reaches 3.69, a relatively small value that confirms the typical small-world navigability: nodes can reach one another via only a limited number of intermediate connections.

Network Topology	Nodes (N)	Edges (E)	Avg. Degree ((k))	Avg. Clustering (C)	Diameter / Path (L)
Empirical Facebook Network	4,039	88,234	43.7	0.6055	8 / 3.69
Erdos-Renyi Random Graph	4,039	88,190	43.7	0.0108	4 / 2.72
Watts-Strogatz Small-World	4,039	88,858	44.0	0.5820	5 / 2.94
Barabasi-Albert (Power-Law)	4,039	88,792	44.0	0.0415	4 / 2.58

3.2: Degree Distribution and Power-Law Test (Corresponding to Figure 2: Log - Log Scatter Plot of Degree Distribution)



The BA network exhibits a characteristic power-law decaying tail. The WS network concentrates degrees within a narrow range without high-degree hubs. The empirical Facebook network displays intermediate behaviour: it possesses heavy-tailed high-degree nodes while maintaining a large share of moderately connected

nodes.

4. Discussion

4.1 Summary of Empirical Findings

Our comparison demonstrates that no single classic generative model can fully replicate all topological features of the real Facebook social network.

The Watts–Strogatz model accurately reproduces the high local clustering coefficient and short path lengths observed in the empirical data, successfully capturing small-world properties. Nevertheless, it fundamentally fails to generate the heavy-tailed, heterogeneous degree distribution found in real social networks, as its degrees remain tightly concentrated around the mean (exhibiting a Poisson-like peak). Conversely, the Barabási–Albert (BA) preferential attachment model properly produces scale-free degree distributions with characteristic power-law tails; yet, its average clustering coefficient ($C=0.0415$) remains critically low and fails to reflect tightly knit local communities. The Erdős–Rényi (ER) random graph underperforms on all fronts: it lacks local clustering ($C=0.0108$) and displays a binomial/Poisson degree distribution, rendering it an inadequate baseline for human online social connections.

4.2 Theoretical Implications

The empirical mismatch of single-mechanism models validates our core research hypothesis. Real-world online social networks simultaneously exhibit small-world clustering and scale-free degree heterogeneity—two distinct topological properties that are artificially segregated across the three baseline models. This implies that simple, isolated generative mechanisms are theoretically insufficient for modeling human social fabric. Unlike abstract mathematical graphs, human social ties are forged through at least two concurrent sociological forces:

4.2.1 Homophily

Individuals preferentially connect with others who share similar traits, interests, or backgrounds, resulting in dense local triads and high clustering.

4.2.2 Cumulative Advantage and Preferential Attachment

Highly active users or prominent public figures naturally

accumulate disproportionately more connections over time, driving the heavy-tailed, power-law degree distribution.

4.3 Limitations

This study contains several limitations. First, the analysis relies on a single Facebook ego-network dataset; conclusions may not generalise to other social platforms such as Twitter or Instagram. Second, we fix network size to match the empirical graph when generating synthetic networks, but parameter calibration is limited and we do not perform systematic parameter sweeping. Third, we only evaluate first-order static topological metrics (clustering coefficient, path length, degree distribution). Higher-order structures such as community partitions and motif statistics are not incorporated into the assessment framework.

4.4 Future Research Directions

Future work can be extended in multiple directions. First, hybrid network generation models that integrate both local triadic closure (to boost clustering) and preferential attachment (to create hubs) should be tested, as they are theoretically more suitable for social network simulation. Second, multi-platform social network datasets can be analysed to compare whether small-world and scale-free coexistence is a universal feature of online social networks. Finally, dynamic network models can be adopted to investigate how social links evolve over time, rather than only examining static network snapshots.

4.5 Concluding Remark

The Facebook network serves as evidence that real human social structures transcend classic single-mechanism network models. Modelling social networks requires theoretical frameworks that unify local clustering mechanisms and global preferential attachment dynamics.

5. Conclusion

This project systematically evaluated three canonical network generative models against the empirical Facebook ego-network dataset using clustering coefficients, average shortest path lengths, and log-log scale degree distributions. The empirical results conclusively confirm that the real Facebook social network

amalgamates small-world structural traits and scale-free degree heterogeneity.

Our findings demonstrate that while the Watts-Strogatz model successfully approximates local clustering and short paths, it fails to generate hub nodes; conversely, the Barabási-Albert model captures power-law degree distributions but fails to simulate dense local communities. The Erdős-Rényi random graph remains entirely inconsistent with observed structural features.

Ultimately, real human social networks cannot be fully captured by any single traditional generation mechanism, pointing the field toward advanced hybrid modeling as the necessary next frontier.